1 Explanation certificate: heart-s2-c2

This section is the explanation certificate of the heart-s2-c2 network. It is rendered from the canonical JSON record `heart-s2-c2.json` (schema 1.4.0, SHA-256 `1276e9a13f62dbc0...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 1 summarises the network and its coverage.

Dataset	Heart Disease (binary, prepared)
Experiment	10seed-heart-s2, seed 2
Genome	`d4bbc960-816a-4781-8979-fdd4165debf2` (NEAT key 9685, generation 39)
Explanation	`3531d96c-aa00-4141-863a-2c85155afe8f` (31 operations)
Train AUC / accuracy	0.9403 / 0.8967 (n = 242)
Test AUC / accuracy	0.9009 / 0.8197 (n = 61)
Output activation	sigmoid
Phenotype	31 nodes (26 inputs, 4 hidden, 1 output), 24 enabled edges
Explained model state	33 nodes, 24 enabled edges
Refactoring operations	split_node 2
Function preservation	max $\|\Delta\,\text{output}\| = 2.22 \times 10^{-16}$ on the training split
Annotations	3 (2 leaf, 1 composition)
Structural coverage	1 (32 of 32 candidate nodes; 8 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`1276e9a13f62dbc0...`

Table 1: Explanation certificate summary: heart-s2-c2.

Drafter's summary. Mostly logistic regression: 16 direct edges (logit SD 3.66; train AUC 0.934 alone, L2 logistic regression 0.9355) plus one consequential nonlinear part to 0.940. The protective gate 3842 = ReLU(1.02 ReLU(0.688 thalach - 0.374 thal:6 + 2.15 [non-anginal pain] - 0.347) - 0.295), read out -0.519, opens for nearly all non-anginal-pain patients, for others only above about +0.8 SD max heart rate, and never with a fixed thallium defect; once open a higher max heart rate lowers risk 3.4x as steeply. A restecg:2 relay is negligible. Unlike the generation-38 heart-s2 record (age/thalach/thal:7 gate, flat-slope gate, oldpeak hinges), age and oldpeak here act only linearly and the gate is keyed on chest-pain type and fixed defect instead.

What resisted explanation. Nothing resisted coverage; only input fan-out needed splitting (thalach, thal:6). Relay 3607 is a step on the binary restecg:2 input, so exactly affine on the data and negligible (read out -0.021). Eight inputs are unconnected (thal:3, thal:4, slope:3, exang:0, exang:1, restecg:0, restecg:1, fbs:0) and listed as zero-effect root entries; exercise angina is ignored entirely. Counter-directional signs reported, not explained: cp:2 +0.26 and fbs:1 -0.22 in the linear core, and thal:6 (fixed defect) -0.30 directly plus closing the protective gate.

1.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes

are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 32%; it adds up most of its inputs independently, and two parts add to that sum: one acts on 104 of 242 training patients through chest pain type, maximum heart rate and the thallium scan, and one through the resting ECG barely changes the predictions. [C29, C37, C41] These add up independently, and these lower the risk: through this sum alone, compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 32% to about 2%, non-anginal chest pain to about 7% and atypical angina to about 14%; being a woman lowers it to about 4% and fasting blood sugar above 120 mg/dL to about 20%; and a higher maximum heart rate lowers it steadily, from about 37% at 118 to about 28% at 175 beats per minute (chest pain type and maximum heart rate also act through the heart rate gate, below). [C04, C12, C16, C08, C18, C20] Each extra vessel coloured by fluoroscopy adds about the same, from about 32% at 0 to about 96% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 81%; and ST depression on exercise, cholesterol and resting blood pressure each raise it steadily, from about 25% at 0 mm to about 55% at 2.8 mm, from about 22% at 193 mg/dL to about 50% at 308 mg/dL, and from about 21% at 110 mm Hg to about 48% at 155 mm Hg. [C01, C02, C03, C07, C09] Through this sum alone a reversible defect on the thallium scan raises it to about 57% (the thallium scan also acts through the heart rate gate, below); older age raises it steadily but only from about 31% at 42 to about 32% at 66 years; and the model ignores chest pain brought on by exercise. [C10, C21, C28] Some rarer results also count (the model has little data on this, n = 15 to 17): a downsloping ST segment raises the risk to about 54%, and through this sum alone a fixed defect on the thallium scan lowers it to about 12%. [C13, C14] This part reads chest pain type, maximum heart rate and the thallium scan, and acts on 104 of 242 training patients, 24 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C37] Across the whole model, non-anginal chest pain lowers the predicted risk, and even more so without a fixed defect on the thallium scan: from about 32% to about 3% then, against about 12% to about 2% with one (the model has little data on this, n = 15). [C33] Across the whole model, a higher maximum heart rate lowers the risk, and even more so with non-anginal chest pain, where it falls steadily from about 7% at 118 to about 2% at 175 beats per minute, flattening out near the bottom because the predicted risk cannot go much lower; for other chest pain types it falls from about 37% to about 26%. [C35] Through this part alone, for most patients a higher maximum heart rate does nothing until about 168 beats per minute, then lowers the risk steadily, but for patients with non-anginal chest pain it lowers it steadily all along: for a typical patient from about 32% at 118 to about 29% at 175 beats per minute (the model has little data on this, n = 14). [C31, C36] This part reads only the resting ECG: signs of left ventricular hypertrophy take a typical patient's predicted risk from about 32% to about 31%, and switching the part off changes any prediction by at most about half a percentage point. [C39, C41]

The whole model. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, chest pain type and sex each move the predicted risk on their own, and two parts add to that sum: a heart rate gate that lowers the risk, and a small step for the resting ECG. [C29, C01, C02, C05, C08] *Evidence.* It is made up of the inputs that feed the score directly and two parts, the heart rate gate and the resting ECG step (each explained below); the score is their sum, which was checked on all 242 training patients. [C29] Chest pain brought on by exercise feeds nothing, so the model ignores it. [C28] Across the whole model, compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 32% to about 2%, non-anginal chest pain to about 3% and atypical angina to about 14%. [C05, C06, C17] Across the whole model, a higher maximum heart rate lowers the risk, for most patients slowly at first, then faster, but for patients with non-anginal chest pain steadily: from about 37% at 118 to

about 26% at 175 beats per minute, against about 28% through the sum alone. [C19, C20] Across the whole model, a reversible defect on the thallium scan raises it to about 57%, and a fixed defect lowers it to about 12% (the model has little data on this, n = 15). [C11, C15]

Heart rate gate. This part lowers the predicted risk for patients with non-anginal chest pain or a high maximum heart rate, unless the thallium scan shows a fixed defect (the model has little data on this, n = 15). [C37, C33] *Evidence.* It acts on 104 of 242 training patients, 24 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C37] Through it, non-anginal chest pain takes a typical patient's predicted risk from about 32% to about 16%, and for the typical patient it acts on a fixed defect on the thallium scan raises it from about 2% to about 4% (the model has little data on this, n = 15). [C30, C32] Through it, for most patients a higher maximum heart rate does nothing until about 168 beats per minute, then lowers the risk steadily, but for patients with non-anginal chest pain it lowers it steadily all along: for a typical patient from about 32% at 118 to about 29% at 175 beats per minute (the model has little data on this, n = 14). [C31, C36] Across the whole model, a higher maximum heart rate lowers the risk more without a fixed defect on the thallium scan, for most patients slowly at first, then faster, but for patients with non-anginal chest pain steadily: from about 37% to about 26%, against about 14% to about 10% with one (the model has little data on this, n = 15). [C34] Its formula reproduces the part exactly on all 242 training patients. [C38]

Resting ECG step. This part barely changes the predictions: signs of left ventricular hypertrophy on the resting ECG take a typical patient's predicted risk from about 32% to about 31%. [C39, C41] *Evidence.* Switching it off changes any training patient's predicted risk by at most about half a percentage point. [C41] Its formula reproduces the part exactly on all 242 training patients. [C42]

Worth checking against what is known. Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 32% to about 96%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C05, C06, C17] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 32% to about 20%). [C18] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk (for a typical patient from about 32% to about 12%), from only 15 training patients. [C15] Worth checking: the model ignores whether exercise brings on chest pain. [C28]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	32% to 96%	log-odds 3.99, OR 54.0; range across patients 3.99 to 3.99	242	train; test unclear (AUC gain 0.037 [-0.038, 0.119], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	32% to 81%	log-odds 2.21, OR 9.14; range across patients 2.21 to 2.21	115, 110	train; test disagrees (AUC gain -0.004 [-0.044, 0.034], n=61)	yes	on the training data only
C03	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: steady; 100% of patients same; verified*	25% to 55%	log-odds 1.31, OR 3.72; range across patients 1.31 to 1.31	242	train; test unclear (AUC gain 0.011 [-0.005, 0.034], n=61)	yes	on the training data only
C04	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 2%	log-odds -3.03, OR 0.05; range across patients -3.03 to -3.03	108, 22	train; test agrees (AUC gain 0.039 [0.009, 0.076], n=61)	yes	clearly
C05	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 2%	log-odds -3.03, OR 0.05; range across patients -3.03 to -3.03	108, 22	train; test agrees (AUC gain 0.057 [0.017, 0.104], n=61)	yes	clearly
C06	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 3%	log-odds -2.53, OR 0.08; range across patients -2.88 to -1.96	108, 70	train; test agrees (AUC gain 0.057 [0.017, 0.104], n=61)	yes	clearly
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	22% to 50%	log-odds 1.28, OR 3.58; range across patients 1.28 to 1.28	242	train; test disagrees (AUC gain -0.011 [-0.039, 0.012], n=61)	yes	on the training data only
C08	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	32% to 4%	log-odds -2.34, OR 0.10; range across patients -2.34 to -2.34	159, 83	train; test unclear (AUC gain 0.027 [-0.01, 0.074], n=61)	yes	on the training data only
C09	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	21% to 48%	log-odds 1.22, OR 3.38; range across patients 1.22 to 1.22	242	train; test unclear (AUC gain 0.009 [-0.013, 0.034], n=61)	yes	on the training data only
C10	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 57%	log-odds 1.04, OR 2.82; range across patients 1.04 to 1.04	137, 90	train; test unclear (AUC gain 0.018 [-0.012, 0.053], n=61)	yes	on the training data only
C11	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 57%	log-odds 1.04, OR 2.82; range across patients 1.04 to 1.04	137, 90	train; test unclear (AUC gain 0.018 [-0.012, 0.053], n=61)	yes	on the training data only
C12	direction: chest pain type:	32% to 7%	log-odds -1.75, OR	108, 70	train; test agrees	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C13	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	32% to 54%	log-odds 0.94, OR 2.56; range across patients 0.94 to 0.94	115, 17 (small n)	train; test disagrees (AUC gain -0.004 [-0.044, 0.034], n=61)	yes	the model has little data on this; on the training data only
C14	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 12%	log-odds -1.24, OR 0.29; range across patients -1.24 to -1.24	137, 15 (small n)	train; test unclear (AUC gain 0.018 [-0.012, 0.053], n=61)	yes	the model has little data on this; on the training data only
C15	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 12%	log-odds -1.00, OR 0.37; range across patients -1.24 to -0.42	137, 15 (small n)	train; test unclear (AUC gain 0.018 [-0.012, 0.053], n=61)	yes	the model has little data on this; on the training data only
C16	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 14%	log-odds -1.05, OR 0.35; range across patients -1.05 to -1.05	108, 42	train; test agrees (AUC gain 0.039 [0.009, 0.076], n=61)	yes	clearly
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 14%	log-odds -1.05, OR 0.35; range across patients -1.05 to -1.05	108, 42	train; test agrees (AUC gain 0.057 [0.017, 0.104], n=61)	yes	clearly
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	32% to 20%	log-odds -0.65, OR 0.52; range across patients -0.65 to -0.65	209, 33	train; test unclear (AUC gain 0.000 [-0.011, 0.013], n=61)	yes	on the training data only
C19	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: accelerating; 67% of patients same; mixed (but steady for patients with non-anginal chest pain)*	37% to 26%	log-odds -0.73, OR 0.48; range across patients -1.32 to -0.51	242	train; test agrees (AUC gain 0.014 [0.002, 0.037], n=61)	yes	clearly
C20	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	37% to 28%	log-odds -0.39, OR 0.68; range across patients -0.39 to -0.39	242	train; test unclear (AUC gain 0.006 [0.000, 0.020], n=61)	yes	on the training data only
C21	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	31% to 32%	log-odds 0.07, OR 1.08; range across patients 0.07 to 0.07	242	train; test disagrees (AUC gain -0.001 [-0.005, 0.000], n=61)	yes	slightly; on the training data only
C22	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (whole model) *shape: categorical; highest*	32% to 31%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	123, 119	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C25	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: steady; 100% of patients same; verified*	25% to 55%	log-odds 1.31, OR 3.72	242	train; test unclear (AUC gain 0.011 [-0.005, 0.034], n=61)	yes	on the training data only
C26	magnitude: chest pain type: no chest pain (asymptomatic) to typical angina, typical patient *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 2%	log-odds -3.03, OR 0.05	108, 22	train; test agrees (AUC gain 0.057 [0.017, 0.104], n=61)	yes	clearly
C27	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	22% to 50%	log-odds 1.28, OR 3.58	242	train; test disagrees (AUC gain -0.011 [-0.039, 0.012], n=61)	yes	on the training data only
C28	null: model ignores exercise-induced angina (exact rule)		max change 0.000 (max |change in log-odds|)	242	train	yes	
C29	structure: made of heart_rate_nonanginal_gate, restecg2_relay and inputs age, sex, cp, trestbps, chol, fbs, thalach, oldpeak, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C30	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	32% to 16%	log-odds -0.79, OR 0.46; range across patients -1.14 to -0.21	172, 70	train; test unclear (AUC gain 0.008 [-0.012, 0.030], n=61)	yes	on the training data only
C31	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 168 beats per minute; 70% of patients same; mixed (but steady for patients with non-anginal chest pain); one point with C36 (median fold 168 beats per minute, typical patient 166 beats per minute, 10th-90th percentile 168 beats per minute to 168 beats per minute)*	32% to 29%	log-odds -0.34, OR 0.71; range across patients -0.93 to -0.12	242	train; test unclear (AUC gain 0.008 [0.000, 0.022], n=61)	yes	slightly; on the training data only
C32	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan (through this part) *shape: categorical; highest to lowest: a fixed defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan*	2% to 4% (typical patient this part acts on)	log-odds 0.24, OR 1.27; range across patients 0.00 to 0.82	227, 15 (small n)	train; test unclear (AUC gain 0.001 [0.000, 0.006], n=61)	yes	slightly; the model has little data on this; on the training data only
C33	interaction: effect of chest pain type (chest pain type other than non-anginal chest pain to non-anginal chest pain) by thallium stress test result: a fixed defect on the thallium scan minus thallium stress test result other than a fixed defect on the thallium scan; stronger in thallium stress test result other than a fixed defect on the thallium scan *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	a fixed defect on the thallium scan: 12% to 2%; thallium stress test result other than a fixed defect on the thallium scan: 32% to 3%	log-odds difference 0.79, OR 2.20	15, 227 (small n)	train; test agrees (AUC gain 0.001 [0.000, 0.006], n=61); group difference on test 0.73 [0.60, 0.85]	yes	the model has little data on this
C34	interaction: effect of maximum heart rate (118 beats per minute to 175 beats per minute) by thallium stress test result: a fixed defect on the thallium scan minus thallium stress test result other than a fixed defect on the thallium scan; stronger in thallium stress test result other than a fixed defect on the thallium scan *shape: accelerating; 70% of patients same; mixed (but steady for patients with non-anginal chest pain)*	a fixed defect on the thallium scan: 14% to 10%; thallium stress test result other than a fixed defect on the thallium scan: 37% to 26%	log-odds difference 0.30, OR 1.35	15, 227 (small n)	train; test agrees (AUC gain 0.001 [0.000, 0.006], n=61); group difference on test 0.34 [0.26, 0.44]	yes	the model has little data on this

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C37	subgroup: who this part acts on: 104 of 242 patients	5% to 2% (typical patient this part acts on (risk without the part, then with it))	share 0.43; log-odds -0.67, OR 0.51; range across patients -1.22 to -0.05	104, 138	train; test unclear (AUC gain 0.018 [0.000, 0.044], n=61)	yes	slightly; on the training data only
C38	structure: its formula reproduces it on 242 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	242	train	yes	
C39	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (through this part) *shape: categorical; highest to lowest: resting ECG other than signs of left ventricular hypertrophy on the resting ECG, signs of left ventricular hypertrophy on the resting ECG*	32% to 31%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	123, 119	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C40	subgroup: who this part acts on: 119 of 242 patients	85% to 84% (typical patient this part acts on (risk without the part, then with it))	share 0.49; log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	119, 123	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	no	slightly; on the training data only
C41	null: this part barely changes predictions (switched off)		max change 0.005 (max |change in predicted risk|)	242	train	yes	
C42	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	

Not supported by the model.

- C40: subgroup claim about resting ECG in part ann_29: not supported by the model (its effect is negligible).

1.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (8):** `-10` (fbs:0), `-12` (restecg:0), `-13` (restecg:1), `-16` (exang:0), `-17` (exang:1), `-21` (slope:3), `-23` (thal:3), `-24` (thal:4).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

1.3 Annotation hierarchy

The hierarchy has 1 root; the global root is heart_disease_logit. Tags (A1, A2, . . .) are used in the figures. ▷ A1: heart_disease_logit (composition, section 1.6.1)

▷ A2: heart_rate_nonanginal_gate (leaf, section 1.6.2)

▷ A3: restecg2_relay (leaf, section 1.6.3)

1.4 Network figures

1.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 242 training rows is 2.22×10^{-16} (tolerance 1.00×10^{-9}).

- #26: split `-15` into `-15_a`, `-15_b`.
- #27: split `-25` into `-25_a`, `-25_b`.

1.6 Annotations

1.6.1 A1: heart_disease_logit

Kind: Composition annotation (depth 0); children A2 (section 1.6.2), A3 (section 1.6.3).

- **Entry nodes:** `-1` (age), `-2` (sex:0), `-3` (sex:1), `-4` (cp:1), `-5` (cp:2), `-7` (cp:4), `-8` (trestbps), `-9` (chol), `-10` (fbs:0), `-11` (fbs:1), `-12` (restecg:0), `-13` (restecg:1), `-15_a` (thalach_a), `-16` (exang:0), `-17` (exang:1), `-18` (oldpeak), `-19` (slope:1), `-20` (slope:2), `-21` (slope:3), `-22` (ca), `-23` (thal:3), `-24` (thal:4), `-25_a` (thal:6_a), `-26` (thal:7).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `-1`→`0` (0.0279), `-2`→`0` (-0.771), `-3`→`0` (0.34), `-4`→`0` (-0.369), `-5`→`0` (0.264), `-7`→`0` (0.869), `-8`→`0` (0.492), `-9`→`0` (0.535), `-11`→`0` (-0.222), `-18`→`0` (0.556), `-19`→`0` (-0.469), `-20`→`0` (0.634), `-22`→`0` (1.78), `-26`→`0` (0.501), `3607`→`0` (-0.0209), `3842`→`0` (-0.519), `-15_a`→`0` (-0.153), `-25_a`→`0` (-0.3).
- **With its descendants:** 33 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 32 candidate nodes.

Hypothesis. Root: heart-disease logit at output 0 (sigmoid). Essentially logistic regression: 16 direct edges (ca +1.78, cp:4 +0.87, sex:0 -0.77, slope:2 +0.63, oldpeak +0.56, chol +0.54, thal:7 +0.50, trestbps +0.49, slope:1 -0.47, cp:1 -0.37, sex:1 +0.34, thal:6 -0.30, cp:2 +0.26, fbs:1 -0.22, thalach -0.15, age +0.03) carry logit SD 3.66 and train AUC 0.9343 alone (L2 logistic regression on all 26 inputs: 0.9355). The two children are added to the logit: heart_rate_nonanginal_gate (x -0.519; +0.006 AUC, the only consequential nonlinearity) and restecg2_relay (x -0.021, negligible). Eight inputs are unconnected, zero effect: thal:3, thal:4, slope:3, exang:0, exang:1, restecg:0, restecg:1, fbs:0 (exercise angina is ignored). Evidence (train, n=242): composed closed form; sensitivity and mean |SHAP| both rank ca first (0.16), then cp:4, sex:0, slope:2; full-model train AUC 0.9403, accuracy 0.897.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= 0.324 + 0.501x_{thal:7} + 0.556x_{oldpeak} + 0.34x_{sex:1} + 0.0279x_{age} \\
&\quad + 0.634x_{slope:2} + 0.264x_{cp:2} + 0.535x_{chol} + 0.869x_{cp:4} + 1.78x_{ca} \\
&\quad + 0.492x_{trestbps} - 0.771x_{sex:0} - 0.153x_{thalach_a} \\
&\quad - 0.0209\,\mathrm{restecg2_relay}(x_{restecg:2}) - 0.369x_{cp:1} \\
&\quad - 0.519\,\mathrm{heart_rate_nonanginal_gate}(x_{cp:3}, x_{thalach_b}, \\
&\quad x_{thal:6_b}) - 0.3x_{thal:6_a} - 0.222x_{fbs:1} - 0.469x_{slope:1}
\end{aligned}
$$

Expanded over the children:

$$y_0 = \sigma(z_0)$$

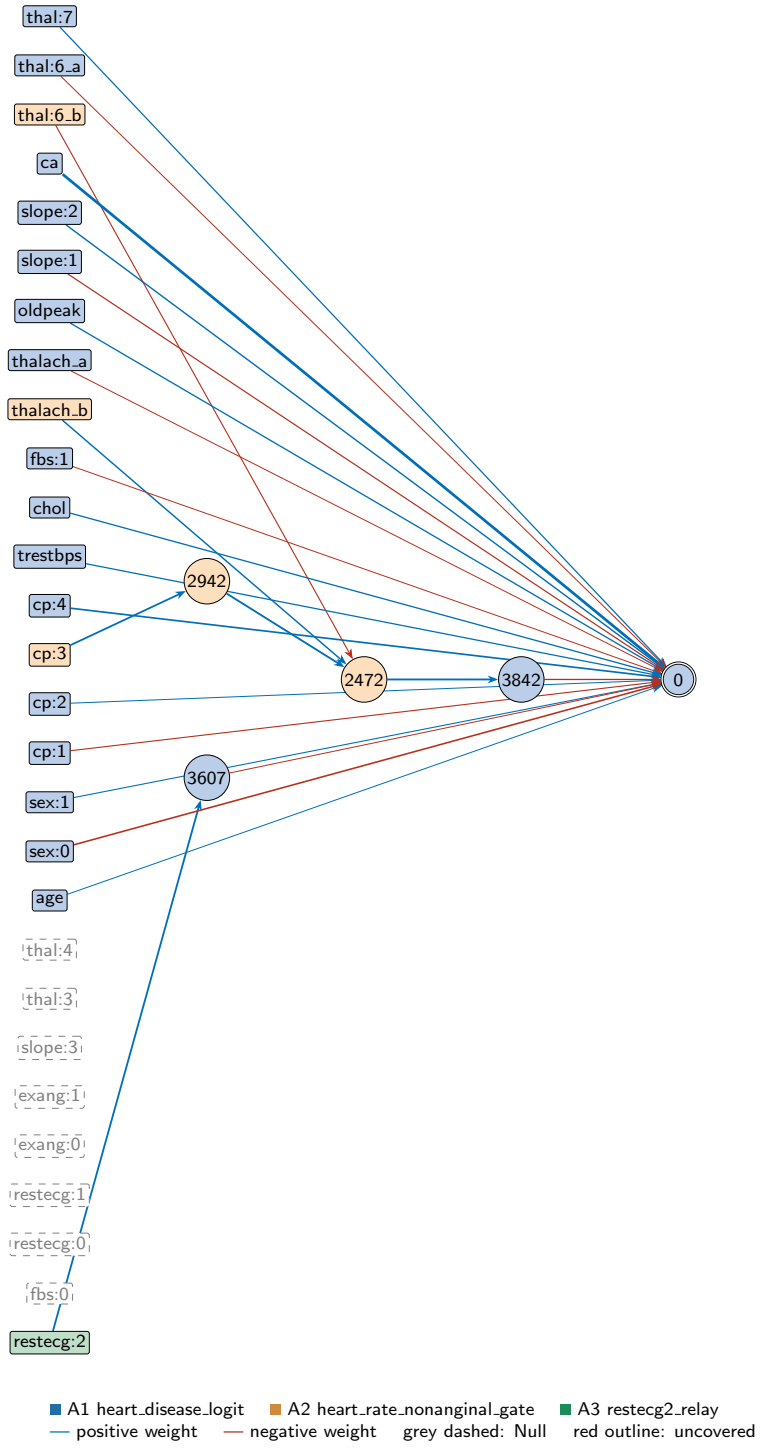


Figure 1: heart-s2-c2: full view of the explained model state.

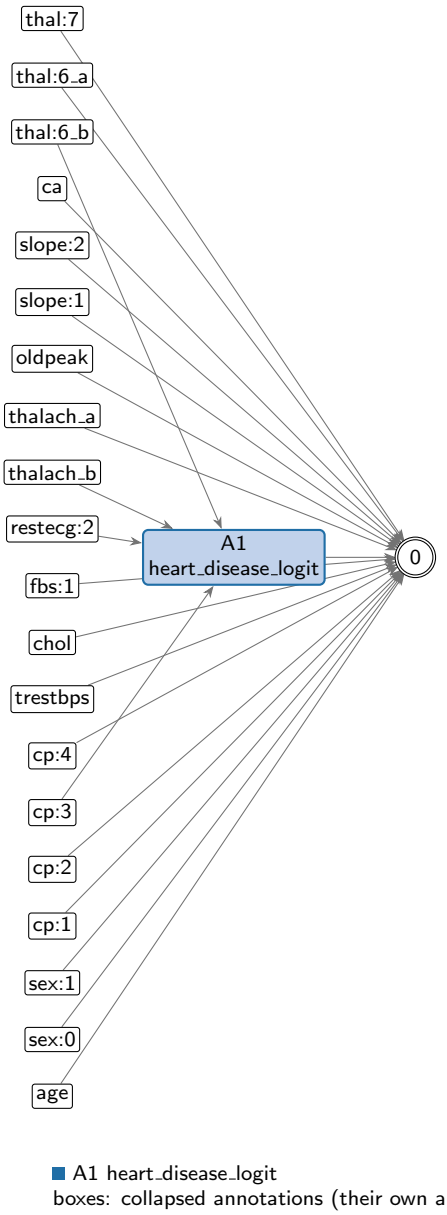


Figure 2: heart-s2-c2: collapsed view at hierarchy level 0 (fully recomposed).

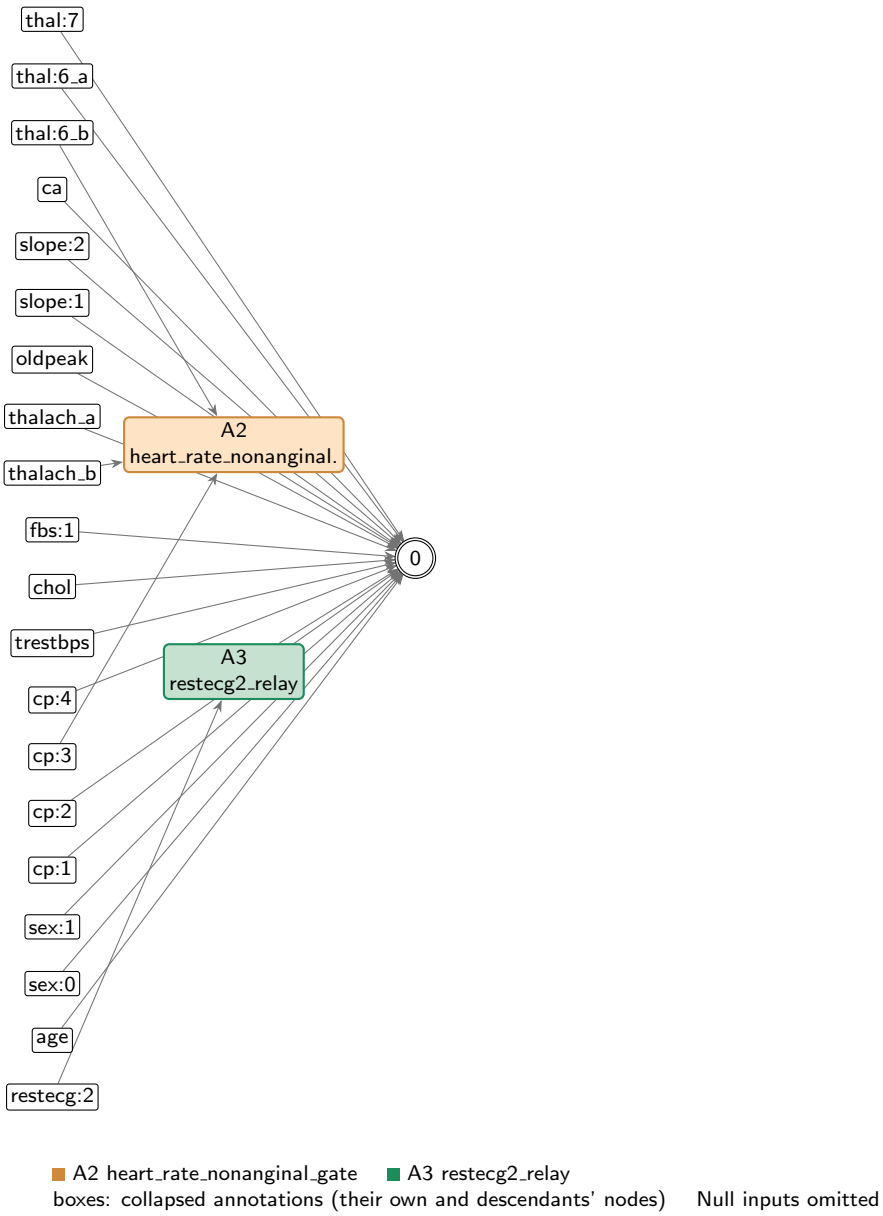


Figure 3: heart-s2-c2: collapsed view at hierarchy level 1.

$$
\begin{aligned}
z_0 = 0.324 &+ 0.501x_{thal:7} + 0.556x_{oldpeak} + 0.34x_{sex:1} + 0.0279x_{age} \\
&+ 0.634x_{slope:2} + 0.264x_{cp:2} + 0.535x_{chol} + 0.869x_{cp:4} + 1.78x_{ca} \\
&+ 0.492x_{trestbps} - 0.771x_{sex:0} - 0.153x_{thalach_a} \\
&- 0.0209\,\mathrm{relu}\big(-0.016 + 0.997x_{restecg:2}\big) - 0.369x_{cp:1} \\
&- 0.519\,\mathrm{relu}\big(-0.295 + 1.02\,\mathrm{relu}\big(-0.347 \\
&+ 0.688x_{thalach_b} + 1.03\,\mathrm{relu}\big(0.571 + 0.968x_{cp:3}\big) \\
&- 0.374x_{thal:6_b}\big)\big) - 0.3x_{thal:6_a} - 0.222x_{fbs:1} - 0.469x_{slope:1}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 8.94×10^{-4}, and the exact function by at most 2.22×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Root closed form (get_formula force=True): direct logistic terms plus the two child functions
- EMPIRICAL, `ablation_study`; data split: train. Train split, full sample; ablations and linear replacements on in-memory copies (no graph edits); SHAP on all 242 train rows

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.2 A2: heart_rate_nonanginal_gate

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-6` (cp:3), `-15_b` (thalach_b), `-25_b` (thal:6_b).
- **Exit nodes:** `3842`.
- **Internal nodes:** `2942`, `2472`.
- **Edges (weight):** `-6`→`2942` (0.968), `-15_b`→`2472` (0.688), `-25_b`→`2472` (-0.374), `2942`→`2472` (1.03), `2472`→`3842` (1.02).
- **Covers** (coverage, own and as part of the hierarchy): 5 candidate nodes.

Hypothesis. Protective heart-rate gate, gated by chest-pain type and thallium. 2942 = ReLU(0.968 cp:3 + 0.571) is a non-anginal-pain indicator (2.09 for the 70 cp:3 patients, 0 otherwise). 3842 = ReLU(1.021*ReLU(0.688 thalach - 0.374 thal:6 + 1.028*2942 - 0.347) - 0.295) is read out -0.519 into the logit. The gate opens once max heart rate passes a threshold set by the other two inputs: about -2.3 SD with non-anginal pain (open for 66 of 68), about +0.78 SD for other pain types without a fixed defect (38 of 159), and never with a fixed thallium defect (0 of 15). Once open, each SD of thalach lowers the logit by 0.36 more (total -0.52 vs -0.15 closed), and non-anginal pain lowers it by about 1.1. Evidence (train, n=242): closed form from get_formula; gate open for 104 patients (disease rate 23% vs 60% when closed); logit contribution SD 0.45; removing it drops train AUC 0.9403 -> 0.9344, a linear refit on cp:3, thalach, thal:6 recovers 0.9378 (R2 0.87), so most of its value is the gating rather than the linear trend. It is the model's only consequential nonlinearity.

Closed form:

$$
\begin{aligned}
y_0 = \mathrm{relu}\big(&-0.295 + 1.02\,\mathrm{relu}\big(-0.347 \\
&+ 0.688x_{thalach_b} + 1.03\,\mathrm{relu}\big(0.571 + 0.968x_{cp:3}\big) \\
&- 0.374x_{thal:6_b}\big)\big)
\end{aligned}
$$

where y_0 is node `3842`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00534, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (heatmap); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.3 A3: restecg2_relay

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-14` (restecg:2).
- **Exit nodes:** `3607`.
- **Internal nodes:** none.
- **Edges (weight):** `-14`→`3607` (0.997).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. restecg:2 relay: 3607 = ReLU(0.997 restecg:2 - 0.016) is 0.998 for the 119 patients with left ventricular hypertrophy on the resting ECG and 0 for the other 123, i.e. exactly affine in the binary input on the data. Read out -0.021, it lowers the logit by 0.02 for LVH: negligible. Evidence (train, n=242): closed form; entry-exit correlation 1.0; logit SD 0.01; removing it changes train AUC 0.9403 -> 0.9401.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.016 + 0.997 x_{restecg:2}\big)$$

where y_0 is node `3607`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 1.39×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (line_plot); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft